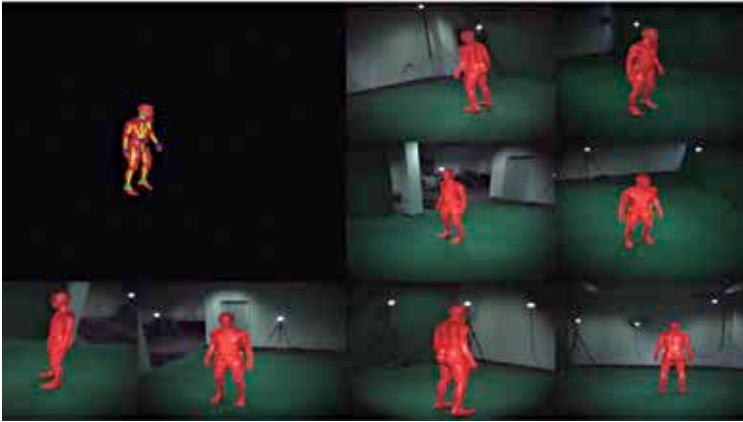




Source: Wikimedia

A robot uses visual data to track the silhouette of a person in the room.

and calculations are required to differing degrees of accuracy. Here's where the software steps in to simplify the entire process.

In the past, control systems would often be implemented mechanically, or directly - for example, the axle of a rotating crank would be connected to a flywheel so as to reduce surges in power, and a governor would be attached that used centrifugal force to rise up, reducing the driving power if the shaft rotated too fast. With the advent of microprocessors and control circuitry, it has become possible to implement far more advanced, measurement-based schema for control.

One classic such method is the PID controller - a Proportional (P), Integral (I) and Differential (D) controller. This type of controller makes use of a term proportional to the difference between current and required values, an integral term of error to check the accumulated errors, and a differential term to control the speed of change (for instance, to avoid jerking).

Another application that is easily implemented using software that supports linear algebra is the Kalman filter. This filter makes use of Gaussian distributions and skillful mathematical manipulation to account for diverse kinds of errors, thereby assuring a high-accuracy prediction.

Sensors and software

Different kinds of sensors often require specific kinds of interfacing with the microcontroller so as to actually produce useful signals. For instance, most Arduino boards are compatible with conventional sensors such as